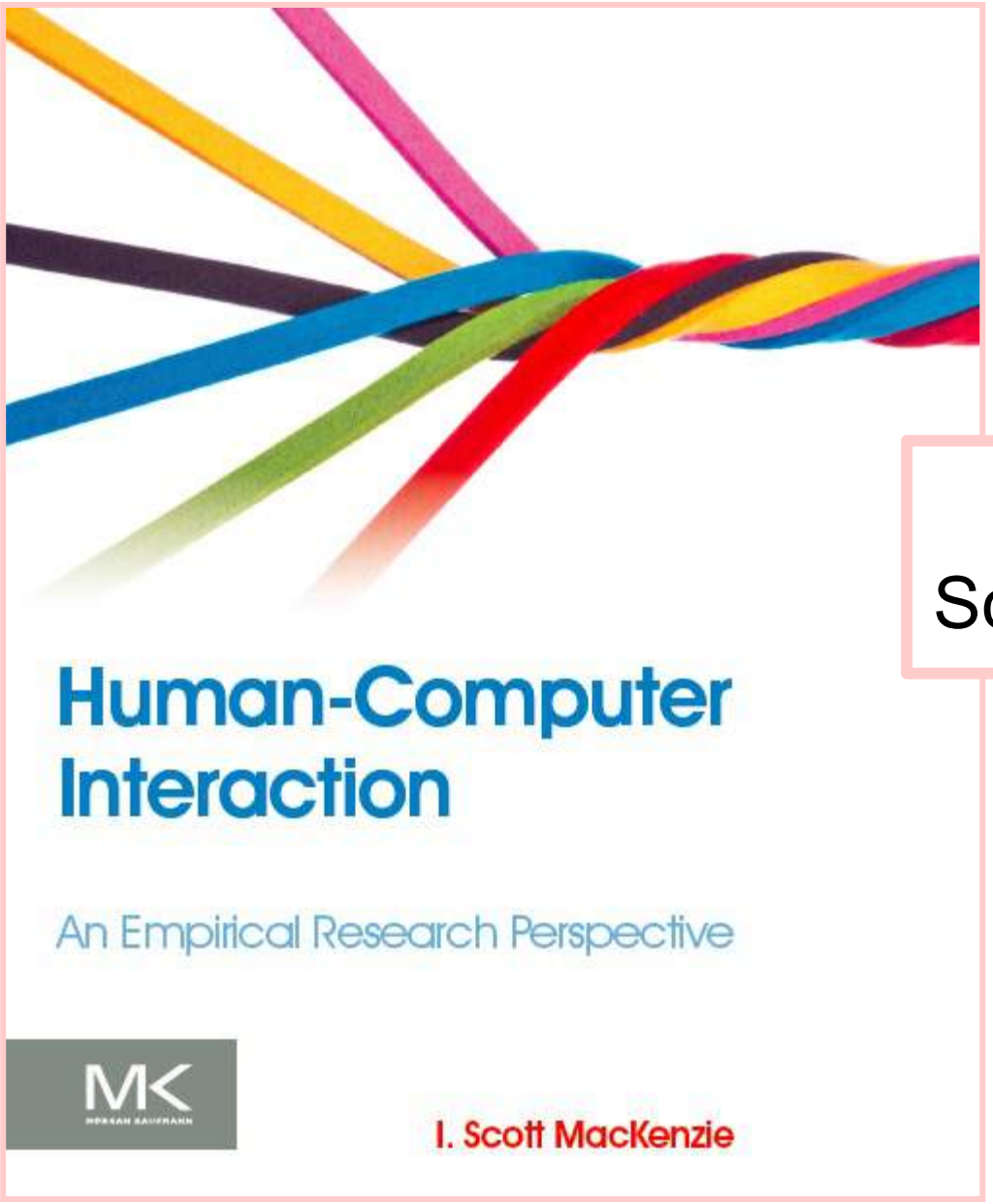



# Chapter 4
# Scientific Foundations

**Longbiao CHEN**
**2018/04 Xiamen University**

# What is Research?

- "Research" means different things to different people
- Often just a word to add weight to an assertion ("*Our research shows that…*")
- ISP television ad:[1]

> "Independent research proves our Internet service is the fastest and most reliable – period."

- Hmm... Is this research available for public scrutiny?
- What about the **independence** of the research?
- *Research* has at least three definitions (next slide)

[1] Aired in Ontario, winter of 2008/2009 (ad for Rogers Communications Inc.).

# Research – Definition #1

- Research is…

Careful or diligent search



- Examples
  - Searching for a bone
  - Searching a computer to find all files modified on a certain date

# Research – Definition #2

- Research is…

Collecting information about a particular subject




- Examples
  - Survey voters in advance of an election
  - Collect information on users' computer habits (e.g., the number of times they … (consulted the manual, clicked the wrong button, uttered an expletive)

# Research – Definition #3

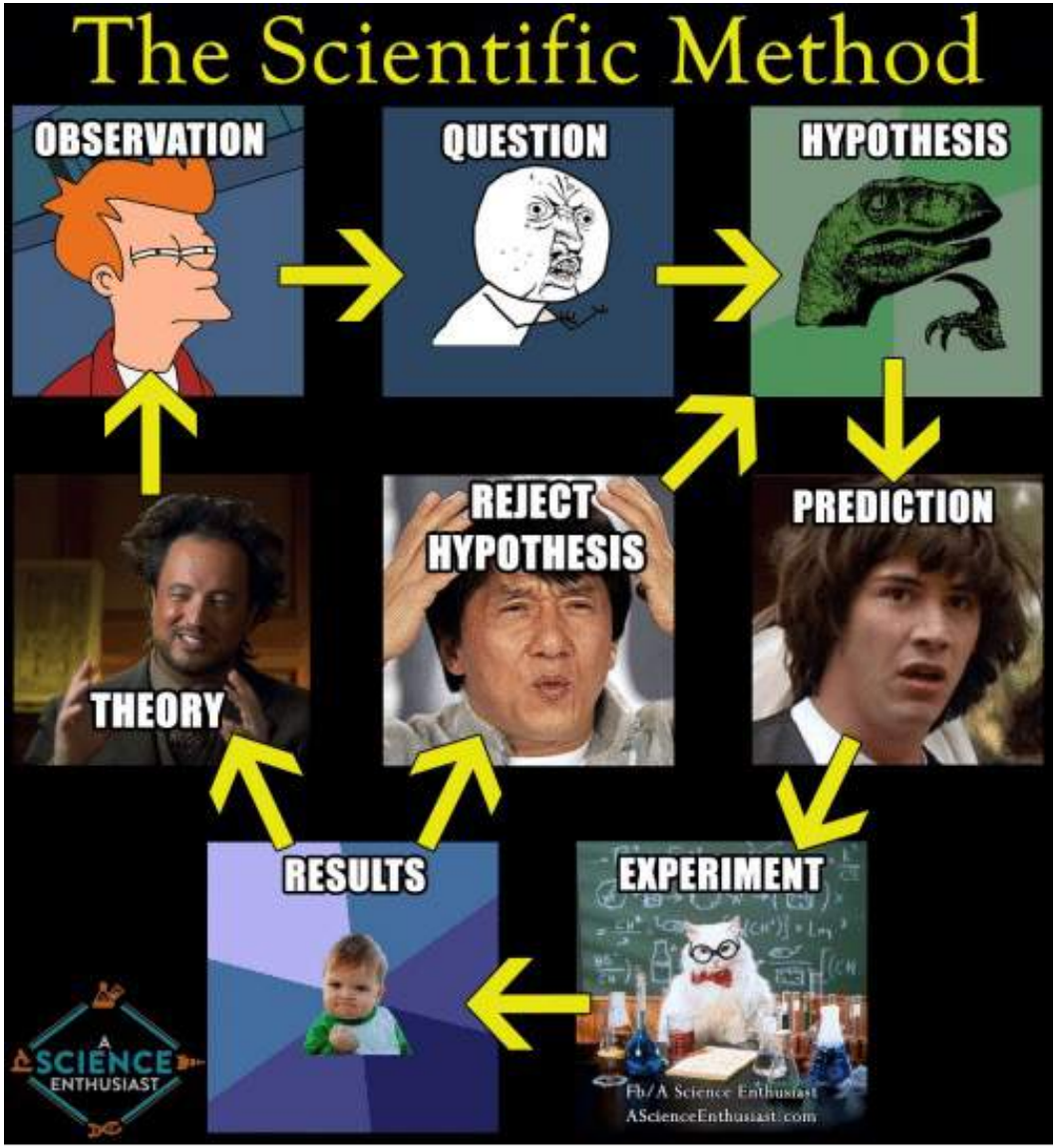
- Research is…

**Investigation or experimentation** aimed at the discovery and interpretation of facts, the revision of accepted theories or laws in light of new facts.



- Example
  - Design and conduct a user study to test whether a new interaction technique improves on an existing interaction technique (that's what this book is all about)

# How to Conduct Research



# Experimentation

- A central activity in HCI research
- An experiment is sometimes called a ***user study***
- Not to be confused with *usability evaluation* (which is different)[1]
- Formal, standardized methodology preferred
  - Brings **consistency** to a body of work
  - Facilitates reviews and **comparisons** between different user studies

[1] MacKenzie, I. S. (2015). User studies and usability evaluations: From research to products. *Proceedings of Graphics Interface 2015*, pp. 1-8. Toronto: Canadian Information Processing Society.

# Facts, Theories, Laws

- **Facts**
  - Building blocks of evidence
  - Evidence is used to test hypotheses (more later)
- **Theory**
  - An hypothesis assumed for the sake of argument
  - A scientifically accepted body of principles that explain phenomena
- **Law**
  - More constraining, more formal, more binding
  - A relationship that is invariable under given conditions
  - HCI involves humans, so laws are of questionable value

# Fitts' Law[1]

- HCI's best known "law"
- Fitts proposed a **model**, not a law
- Fitts' law is a behavioural, predictive, and descriptive model of human motor behaviour
- It is a "law" only in that other researchers took up the label as a celebration of the robustness and importance of Fitts' work

[1] Fitts, P. M. (1954). The information capacity of the human motor system in controlling the amplitude of movement. *Journal of Experimental Psychology, 47*, 381-391.

# Research Must Be Published

- Publication is the final step
- Also an essential step
- *Publish or perish!*
  - Edict for researchers in all fields, and particularly in academia
- Until it is published, research cannot achieve its critical goal:
  - Extend, refine, or revise the existing body of knowledge in the field

# Peer Review

- Research submitted for publication is reviewed by peers – other researchers doing similar research
- Only research meeting a high standard of scrutiny[1] is accepted for publication
  - Are the results novel and useful?
  - Does the evidence support the conclusions?
  - Does the methodology meet the expected standards for the field?
- Accepted research is published and archived
- The final step is complete

1. 仔细检查

# Research Must Be Reproducible

- Research that cannot be replicated is useless
- A high standard or **reproducibility (可重复性)** is essential
- The research write-up must be sufficiently detailed to allow a skilled researcher to replicate the research if he/she desired
- The easiest way to ensure reproducibility is to follow a standardized methodology
- Many great advances in science pertain to methodology (e.g., Louis Pasteur's detailed disclosure of the methodology used in his research on microbiology)
- The most cited research paper is a "method paper"[1] (see Google Scholar for the latest citation count)

[1] Lowry, O. H., Rosenbrough, N. J., Farr, A. L., & Randall, R. J. (1951). Protein measurement with the Folin phenol reagent. *Journal of Biological Chemistry, 193*, 265-275.

# Research vs. Engineering vs. Design

- Researchers often work closely with engineers and designers, but the skills each bring are different
- Engineers and designers are in the business of building things, bringing together the best in *form* (design emphasis) and *function* (engineering emphasis)
- One can image that there is a certain tension, even trade-off, between form and function

# Form Trumping Function

- The photo below shows part of a laptop computer
- The form is elegant – smooth, shiny, metallic
- The touchpad design (or is in engineering?) has a problem
- No tactile sense at the sides of the touchpad



The fix

# Duct Tape To: The Rescue



**A true story!**

# Example: Apple *iPhone* (2007)

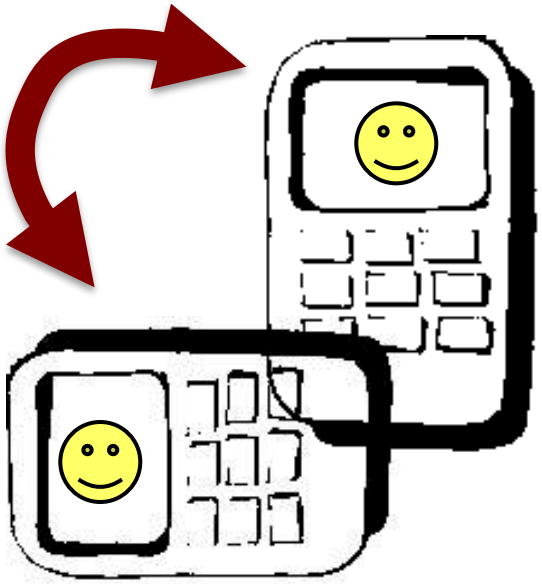



*iPhone* Gestures:

- Tilt
- Multitouch
- Flick

# Tilt

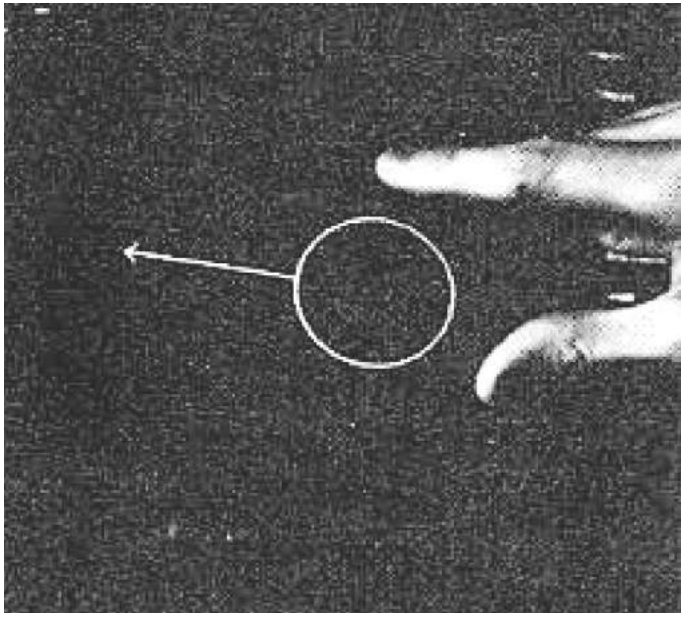
- Research on tilt as an interaction primitive dates at least to 1998[1]



[1] Harrison, B., Fishkin, K. P., Gujar, A., Mochon, C., & Want, R. (1998). Squeeze me, hold me, tilt me! An exploration of manipulative user interfaces. *Proc CHI '98*, 17-24, New York: ACM.

# Multitouch

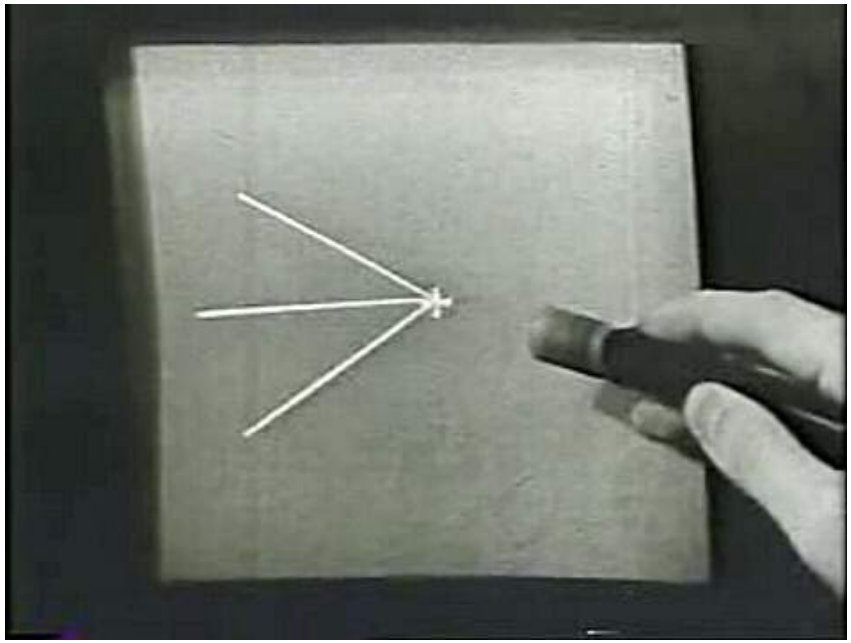
- Research on multitouch as an interaction primitive dates at least to 1978[1]



[1] Herot, C. F., & Weinzapfel, G. (1978). One-point touch input of vector information for computer displays. *Proceedings of SIGGRAPH 1978*, 210-216, New York: ACM.

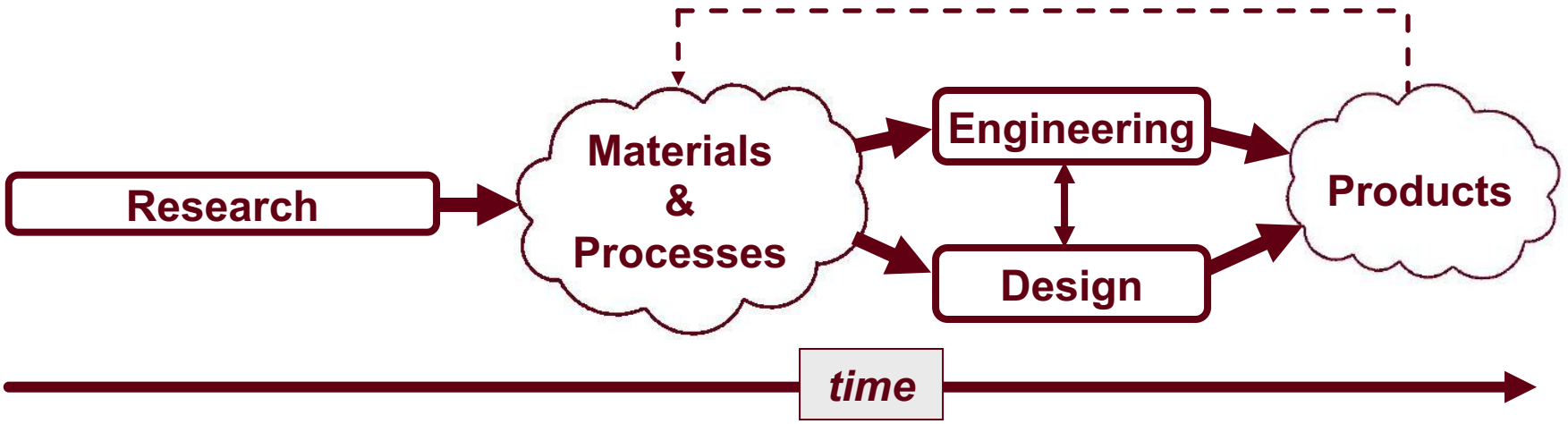
# Flick (缩放)

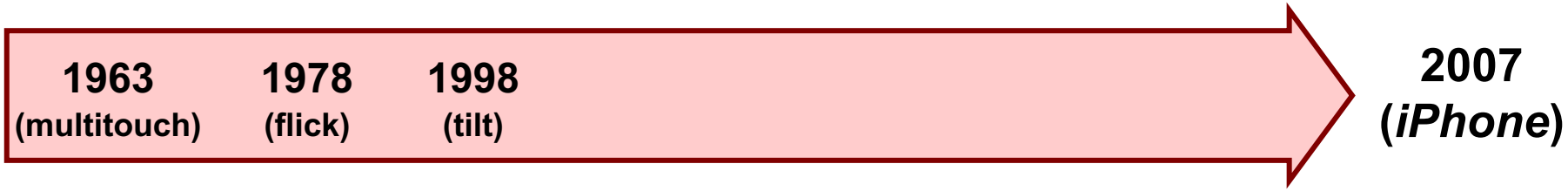
- Research on flick as an interaction primitive dates at least to 1963[1]



[1] Sutherland, I. E. (1963). Sketchpad: A man-machine graphical communication system. *Proceedings of the AFIPS Spring Joint Computer Conference*, 329-346, New York: ACM.

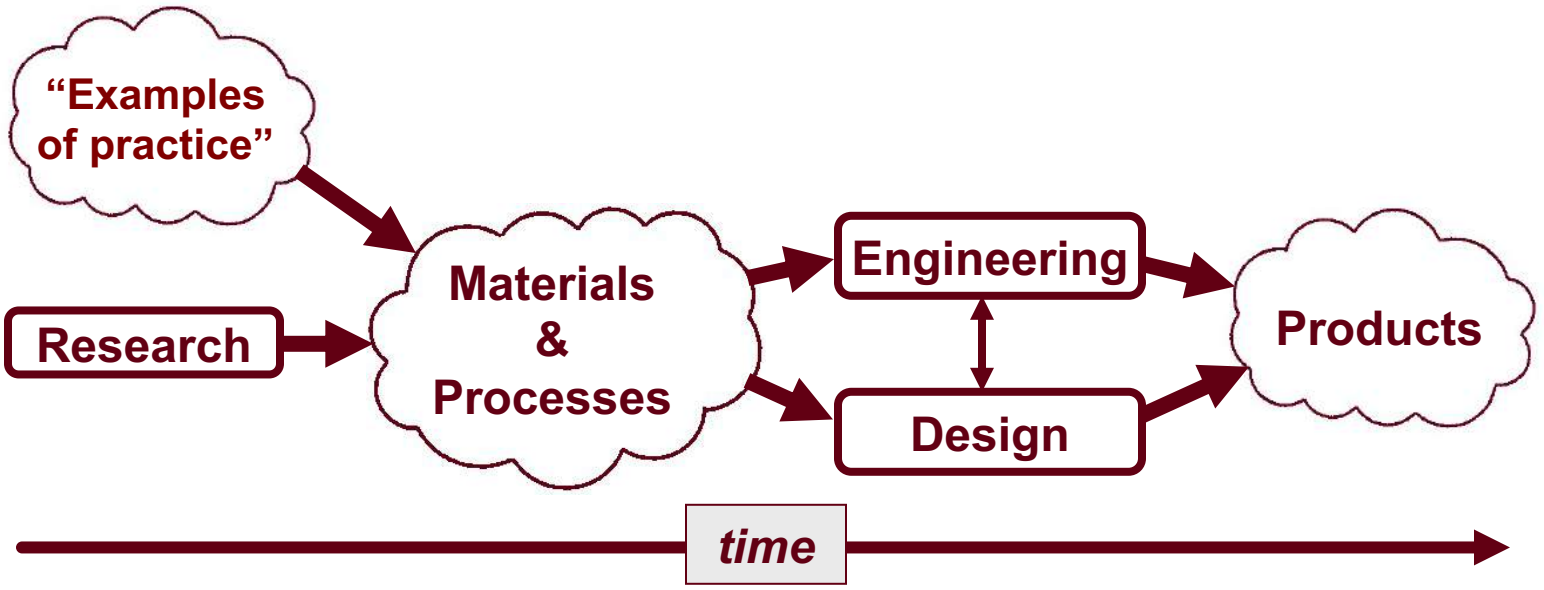
# So…


Research
Materials
&
Processes
Engineering
Design
Products
time


1963
(multitouch)
1978
(flick)
1998
(tilt)
2007
(iPhone)

# On Materials and Processes

- Gaver and Bowers also comment on the materials and processes they use in design:
  - *It was by looking at specific examples of practice that we found guidance for our work.*



# “Empirical” Research

- Empirical:
  - Originating in or based on observation or experience
  - Relying on experience or observation alone without due regard for system or theory (i.e., don’t be blinded by pre-conceptions)
- Example: **Nicolas Copernicus**[1] (1473-1543)
  - Prevailing system or theory: celestial bodies revolved around the earth
  - Copernicus made astronomical observations that cut against this view
  - Result: heliocentric cosmology[2] (the earth and planets revolve around the sun)

1 哥白尼　　2 日心说

# “Empirical” Research (2)

- Empirical: (by another definition)
  - Capable of being verified or disproved by observation or experiment
- HCI research
  - Framed by **hypotheses**
  - Methodology to **test** hypotheses
  - **Experiments** (aka user studies) are the vehicle
  - Hypotheses must be sufficiently narrow and clear to allow for verification or disproval (more on this later; see Research Questions)

# Research Methods

- Observational method
- Experimental method
- Correlational method

# Observational Method

- Example techniques:
  - Interviews, field investigations, contextual inquiries, case studies, field studies, focus groups, think aloud protocols, story telling, walkthroughs, cultural probes, etc.
- HCI approach: *usability evaluation*
- Focus on qualitative assessments (cf. quantitative)
- Relevance vs. precision
  - High in relevance (behaviours studied in a natural setting)
  - Low in precision (lacks control available in a laboratory)
- Goal: discover and explain reasons underlying human behaviour (*why* or *how*, as opposed to *what*, *where*, or *when*)

# Experimental Method

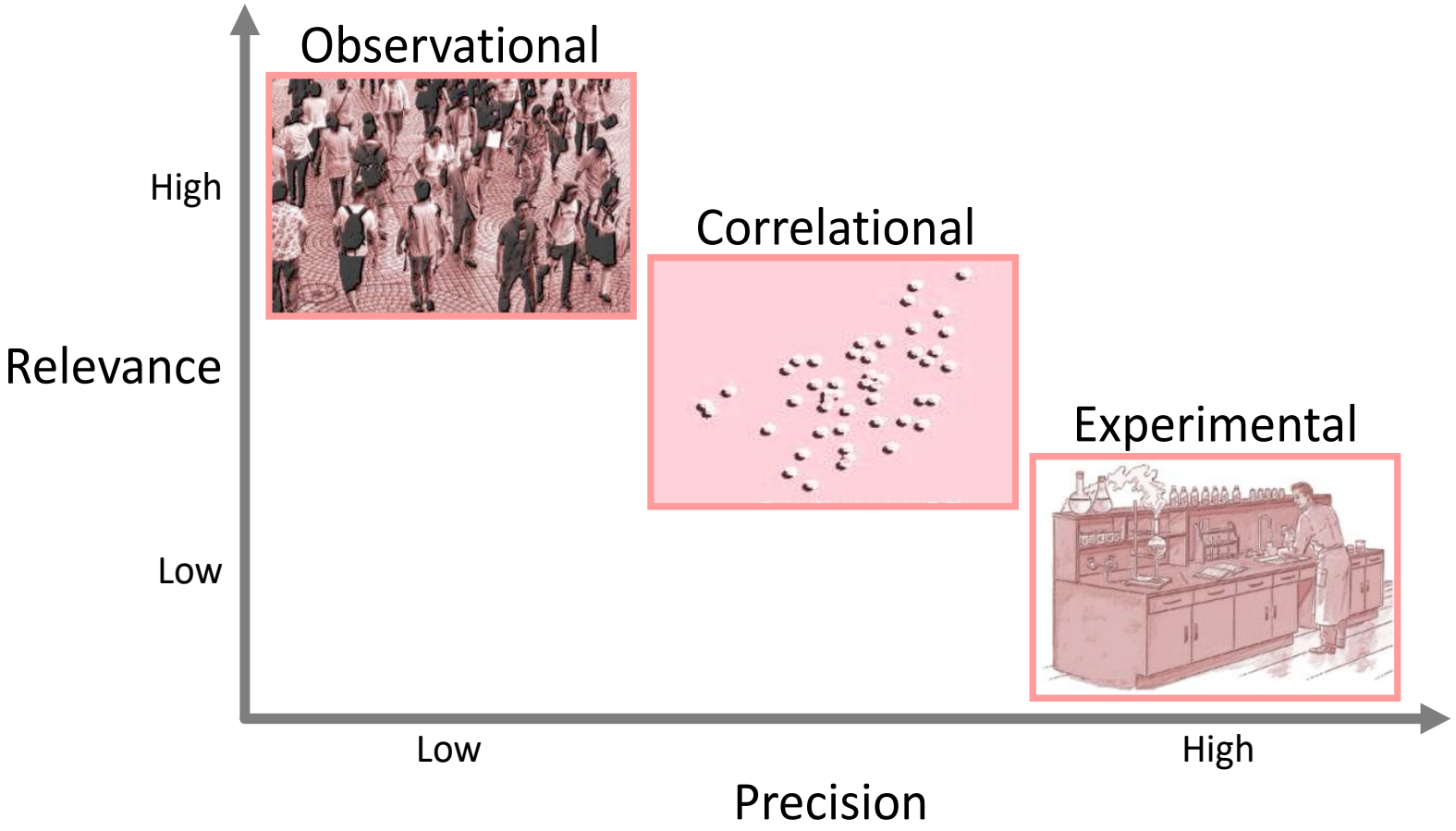
- Aka ***scientific method***
- Controlled experiments conducted in lab setting
- HCI approach: *user study*
- Relevance vs. precision
  - Low in *relevance* (artificial environment)
  - High in *precision* (extraneous behaviours easy to control)
- At least two variables:
  - *Manipulated variable (*aka *independent variable)*
  - *Response variable* (aka *dependent variable*)
- Cause-and-effect conclusions possible (changes in the manipulated variable *caused* changes in the response variable)

# Correlational Method

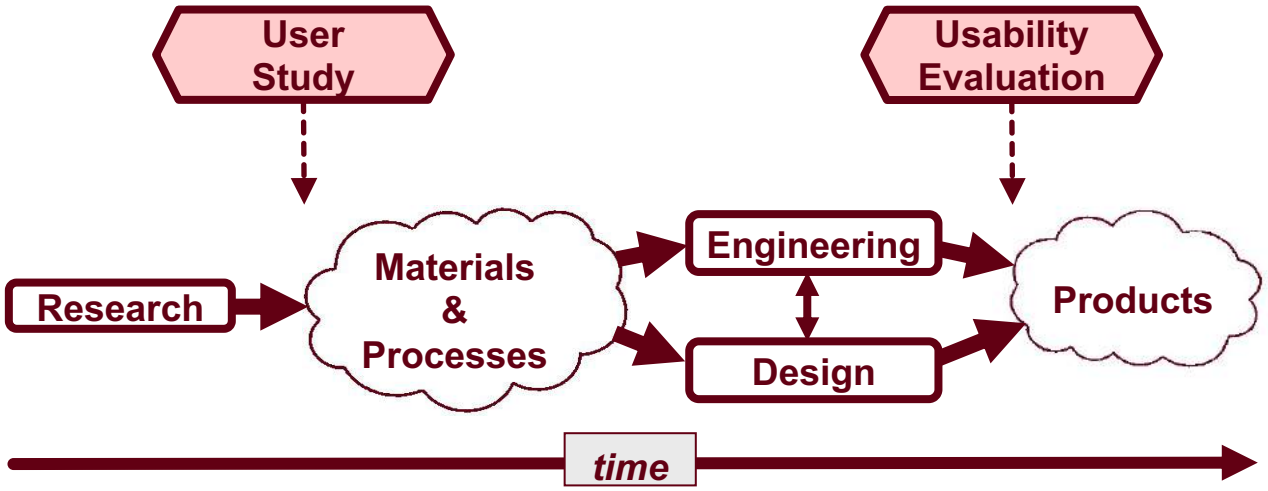
- Look for relationships between variables
- Observations made, data collected
  - Example: Are user's privacy settings while social networking related to their age, gender, level of education, employment status, or income?
- Non-experimental
  - Interviews, on-line surveys, questionnaires, etc.
- Balance between relevance and precision (some quantification, observations not in lab)
- Cause-and-effect conclusions not possible

# Summary

(research methods, relevance, precision)



# User Study vs. Usability Evaluation



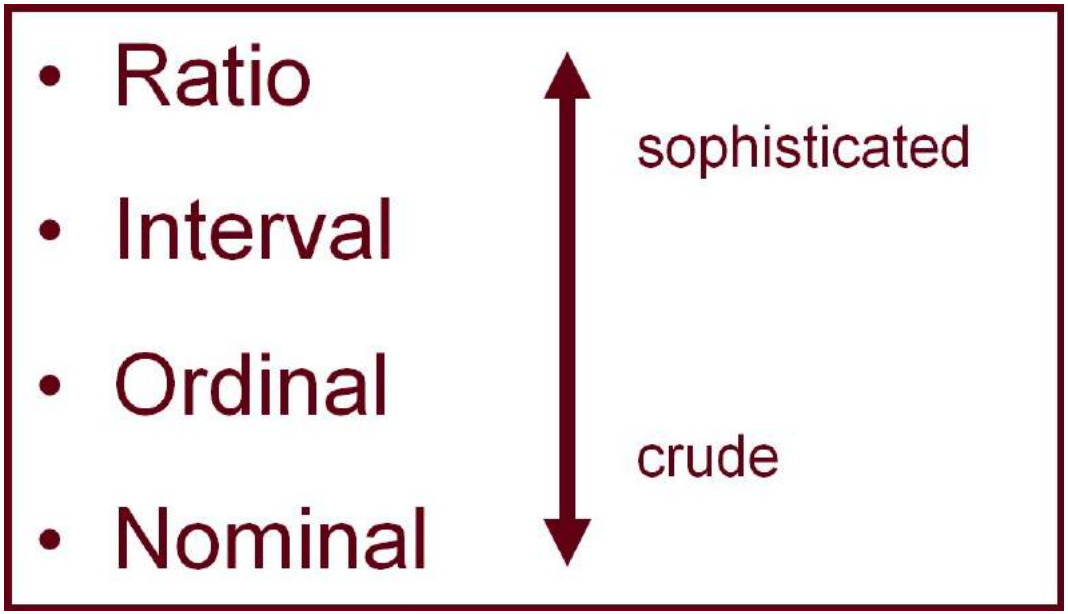

| Feature | User Study | Usability Evaluation |
|---|---|---|
| Manipulated variable(s)? | Yes | No |
| Research method | Experimental | Observational |
| Place in timeline | Early | Late |
| Level of inquiry | Low | High |

MacKenzie, I. S. (2015). User studies and usability evaluations: From research to products. *Proceedings of Graphics Interface 2015 - GI 2015*, pp. 1-8. Toronto: Canadian Information Processing Society.

# Observe and Measure

- **Foundation** of empirical research
- Observation is the starting point; observations are made…
  - By the apparatus (设备)
  - By a human observer
- Manual observation
  - Log sheet, notebooks
  - Screen capture, photographs, videos, etc.
- Measurement
  - With measurement, anecdotes (奇闻轶事) turn to empirical **evidence**
  - "*When you cannot measure, your knowledge is of a meager and unsatisfactory kind*" (Kelvin)

# Scales of Measurement



# Nominal Data

- Nominal data (aka ***categorical*** *data*) are arbitrary codes assigned to attributes; e.g.,
  - 1 = male, 2 = female
  - 1 = mouse, 2 = touchpad, 3 = pointing stick
- The code needn't be a number; i.e.,
  - M = male, F = female
- Note: the statistical **mean cannot** be computed on nominal data
- Usually it is the *count* that is important
  - "Are females or males more likely to…"
  - "Do left handers or right handers have more difficulty with…"
  - Note: The count itself is a ratio-scale measurement
- (example on next slide)

# Nominal Data – HCI Example

- Task: Observe students "on the move" on university campus
- Code and count students by…
  - Gender (male, female)
  - Mobile phone usage (not using, using)

| Gender | Mobile Phone Usage | | Total | % |
|---|---|---|---|---|
| | Not Using | Using | | |
| Male | 683 | 98 | 781 | 51.1% |
| Female | 644 | 102 | 746 | 48.9% |
| Total | 1327 | 200 | 1527 | |
| % | 86.9% | 13.1% | | |

Real Data !

# Ordinal Data

- Ordinal data associate an **order or rank** to an attribute
- The attribute is any characteristic or circumstance of interest; e.g.,
  - Users try three GPS systems for a period of time, then rank them: 1st, 2nd, 3rd choice
- More sophisticated than nominal data
  - Comparisons of "greater than" or "less than" possible
- (example on next slide)

# Ordinal Data – HCI Example

How many email messages do you receive each day?

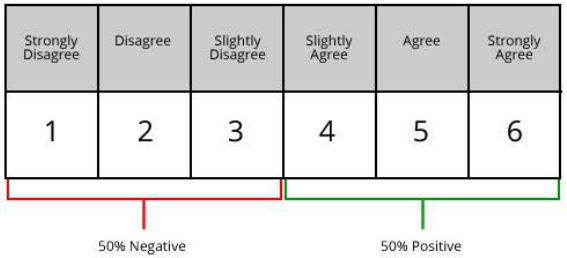
1. None (I don't use email)
2. 1-5 per day
3. 6-25 per day
4. 26-100 per day
5. More than 100 per day

# Interval Data

- Equal distances **between** adjacent values
- **But, no absolute zero**
- Classic example: temperature (°F, °C)
- Statistical mean possible
  - E.g., the mean midday temperature during July
- Ratios not possible
  - Cannot say 10 °C is twice 5 °C

# Interval Data – HCI Example

- Questionnaires often ask for a level of agreement to a statement
- Responses on a Likert scale
- Likert scale characteristics:
  1. Statement soliciting level of agreement
  2. Responses are symmetric about a neutral middle value
  3. Gradations between responses are equal (more-or-less)
- Assuming "equal gradations", the statistical mean is valid (and related statistical tests are possible)
- Likert scale example → (next slide)






# Interval Data – HCI Example (2)

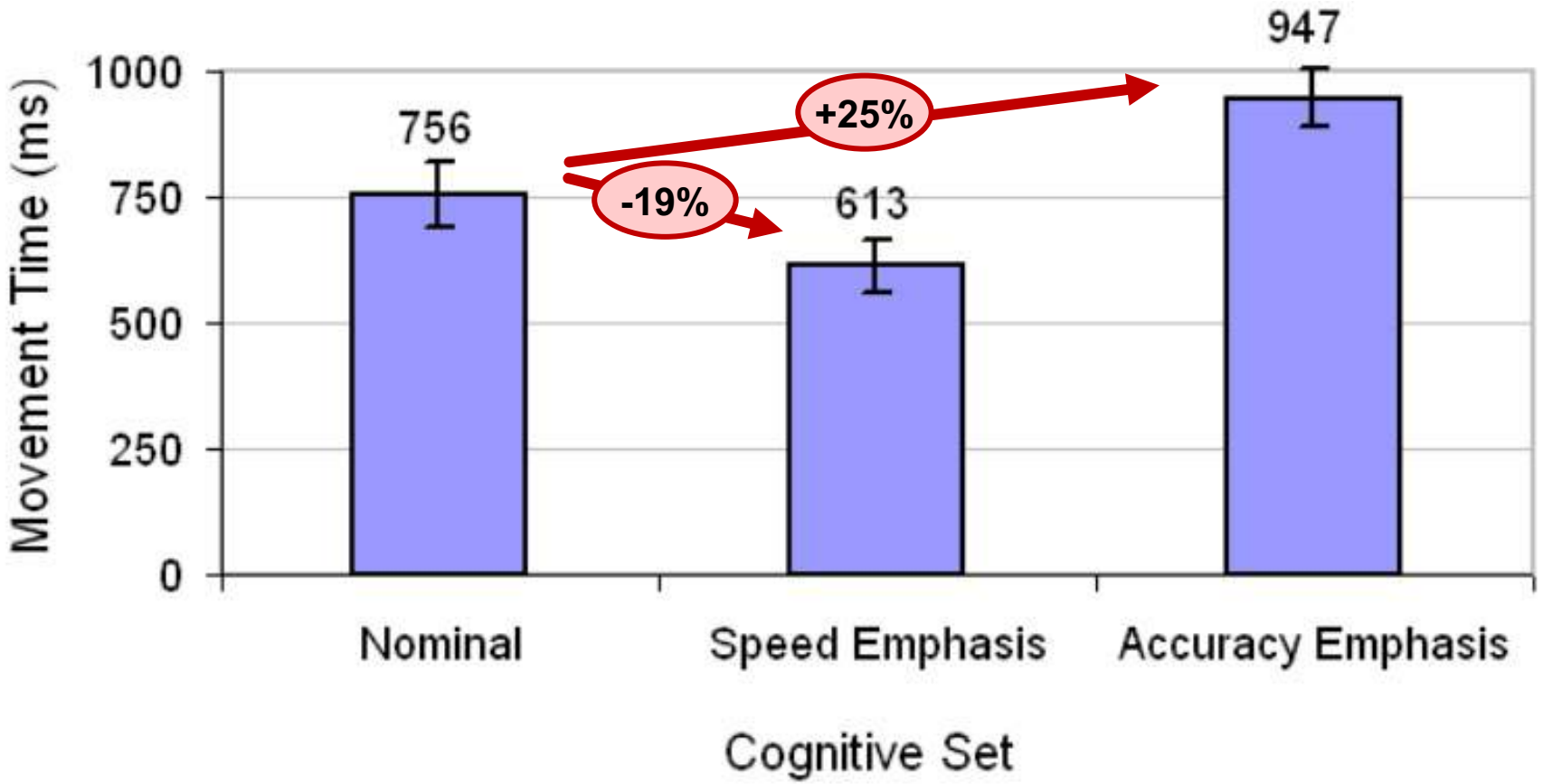
Please indicate your level of agreement with the following statements.

| | Strongly disagree | Mildly disagree | Neutral | Mildly agree | Strongly agree |
|---|---|---|---|---|---|
| It is safe to talk on a mobile phone while driving. | 1 | 2 | 3 | 4 | 5 |
| It is safe to read a text message on a mobile phone while driving. | 1 | 2 | 3 | 4 | 5 |
| It is safe to compose a text message on a mobile phone while driving. | 1 | 2 | 3 | 4 | 5 |

# Ratio Data

- Most sophisticated of the four scales of measurement
- Preferred scale of measurement
- **Absolute zero, therefore many calculations possible**
- Summaries and comparisons are strengthened
- A "count" is a ratio-scale measurement
  - E.g., "time" (the number of seconds to complete a task)
- Enhance counts by adding further ratios where possible
  - Facilitates comparisons
  - Example – a 10-word phrase was entered in 30 seconds
    - Bad: $t$ = 30 seconds
    - **Good: Entry rate = 10 / 0.5 = 20 wpm**

# Ratio Data – HCI Example[1]




[1] MacKenzie, I. S., & Isokoski, P. (2008). Fitts' throughput and the speed-accuracy tradeoff. *Proc CHI 2008*, 1633-1636, New York: ACM.

# Research Questions

- We conduct empirical research to answer (and raise!) questions about UI designs or interaction techniques
- Consider the following questions:
  - Is it **viable** (可行的)?
  - Is it **better** than current practice?
  - Which design alternative is **best**?
  - What are the **performance** limits?
  - What are the **weaknesses**?
  - Does it work well for **novices** (新手)?
  - How much **practice** is required?

# Testable Research Questions

- Preceding questions, while unquestionably relevant, are not testable
- Try to re-cast as testable questions (even though the new question may appear less important)
- Scenario…
  - You have invented a new text entry technique for touchscreen mobile phones, and you think it's pretty good. In fact, you think it is better than the Qwerty soft keyboard (QSK). You decide to undertake a program of empirical enquiry to evaluate your invention. What are your research questions?

# Research Questions (2)

- Very weak

  *Is the new technique any good?*

- Weak

  *Is the new technique better than QSK?*

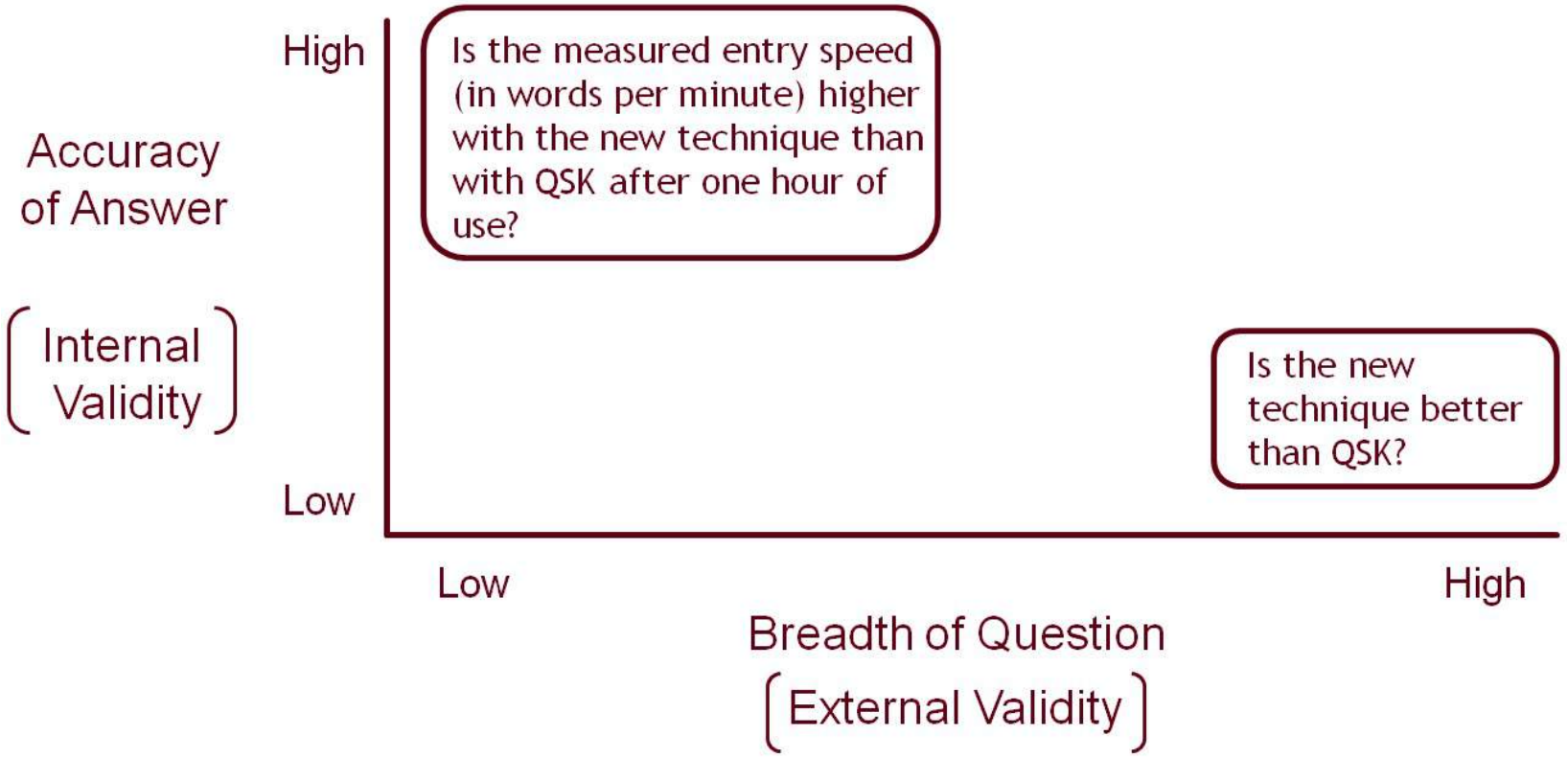
- Better

  *Is the new technique faster than QSK?*

- Better still

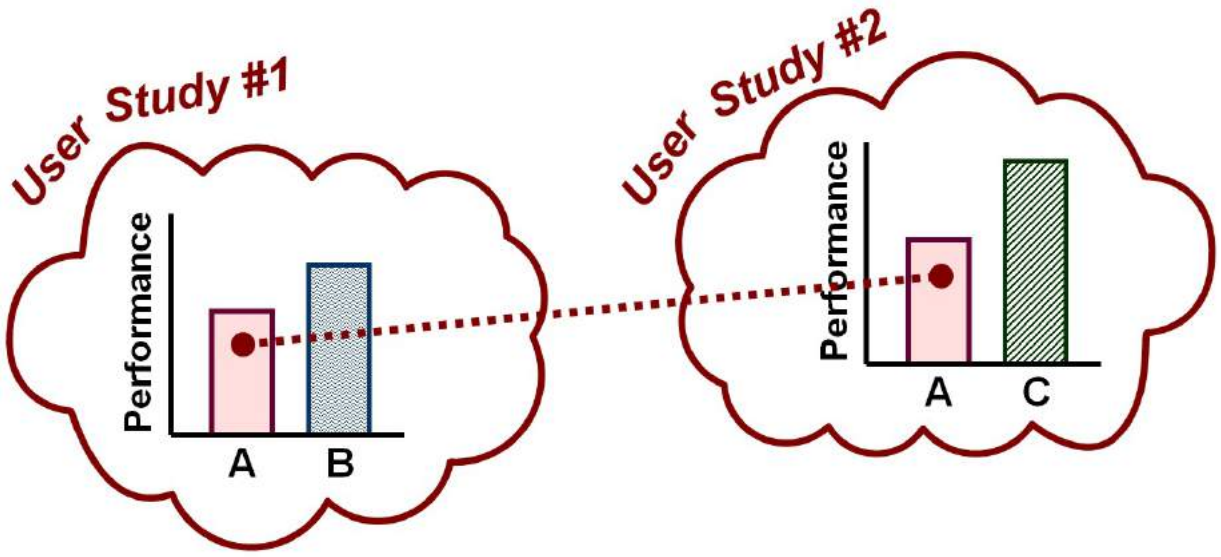
  *Is the measured entry speed (in words per minute) higher for the new technique than for QSK after one hour of use?*

# A Tradeoff


High
Low
Accuracy of Answer
Internal Validity
Is the measured entry speed (in words per minute) higher with the new technique than with QSK after one hour of use?
Is the new technique better than QSK?
Low
High
Breadth of Question
External Validity

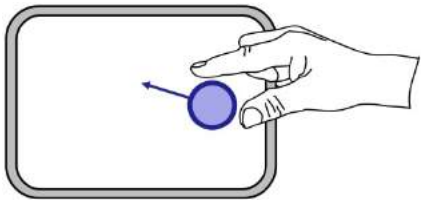
# Comparative Evaluations

- Preferable to do a comparative evaluation rather than one-of
- More **insightful** results obtained
- If one condition is a base line; comparisons possible between studies (assuming similar methodology)



# Research Topics

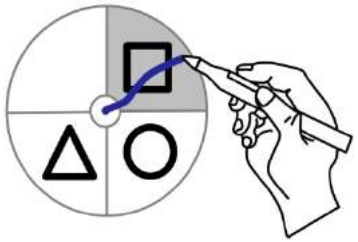
- Finding a research topic is a challenge (for students… and for seasoned researchers too!)









- Four tips:
  1. Think small
  2. Replicate
  3. Know the literature
  4. Think inside the box

# Tip #1 - Think Small

- Looking for big idea?
- Advice: Forget it (besides, it isn't necessary)
- Research questions are small, narrowly focused
- Pursue several small, related research topics and before you know it, a dissertation topic is formed

# Tip #2 - Replicate

- Where's the research in simply replicating what was done before?
- Of course, there is no research in replication, but the trick is in the **path to replicating**
- Replicating prior research is a lot of work
- Along the way, you will undoubtedly discover small and novel **improvements** – things to try
- A little tweak here, a small modification there
- BTW, you might not find a novel idea until well into the process (perhaps afterward!)

# Tip #3 – Know The Literature

- Whatever topic interests you, read the literature
- E.g., *social networking*, *gaming*
- If too broad, narrow (e.g., *privacy settings* in social networking, *avatars* in gaming)
- Read papers, open a spreadsheet, tabulate variables in the methodology and the findings
- **Chaotic at first, order and shape will emerge** (eventually)
- With some **luck** (and further study) a research topics will emerge

# Example Table[1]

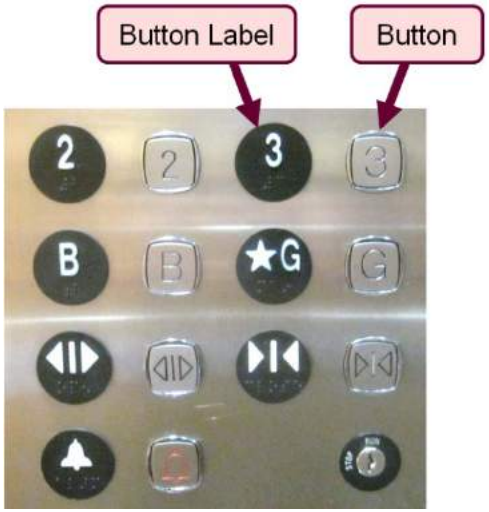
| Study (1st author) | Number of Keys[a] | Direct/ Indirect | Scanning | Number of Participants | Speed[b] (wpm) | Notes |
|---|---|---|---|---|---|---|
| Bellman [2] | 5 | Indirect | No | 11 | 11 | 4 cursors keys + SELECT key. Error rates not reported. No error correction method. |
| Dunlop [4] | 4 | Direct | No | 12 | 8.90 | 4 letter keys + SPACE key. Error rates reported as "very low." |
| Dunlop [5] | 4 | Direct | No | 20 | 12 | 4 letter keys + 1 key for SPACE/NEXT. Error rates not reported. No error correction method. |
| Tanaka-Ishii [25 | 3 | Direct | No | 8 | 12+ | 4 letters keys + 4 keys for editing, and selecting. 5 hours training. Error rates not reported. Errors corrected using CLEAR key. |
| Gong [7] | 3 | Direct | No | 32 | 8.01 | 3 letter keys + two additional keys. Error rate = 2.1%. Errors corrected using DELETE key. |
| MacKenzie [16] | 3 | Indirect | No | 10 | 9.61 | 2 cursor keys + SELECT key. Error rate = 2.2%. No error correction method. |
| Baljko [1] | 2 | Indirect | Yes | 12 | 3.08 | 1 SELECT key + BACKSPACE key. 43 virtual keys. RC scanning. Same phrase entered 4 times. Error rate = 18.5%. Scanning interval = 750 ms. |
| Simpson [24] | 1 | Indirect | Yes | 4 | 4.48 | 1 SELECT key. 26 virtual keys. RC scanning. Excluded trials with selection errors or missed selections. No error correction. Scanning interval = 525 ms at end of study. |
| Koester [10] | 1 | Indirect | Yes | 3 | 7.2 | 1 SELECT key. 33 virtual keys. RC scanning with word prediction. Dictionary size not given. Virtual BACKSPACE key. 10 blocks of trials. Error rates not reported. Included trials with selection errors or missed selections. Fastest participant: 8.4 wpm. |

[a] For "direct" entry, the value is the number of letter keys. For "indirect" entry, the value is the total number of keys.
[b] The entry speed cited is the highest of the values reported in each source, taken from the last block if multiple blocks.

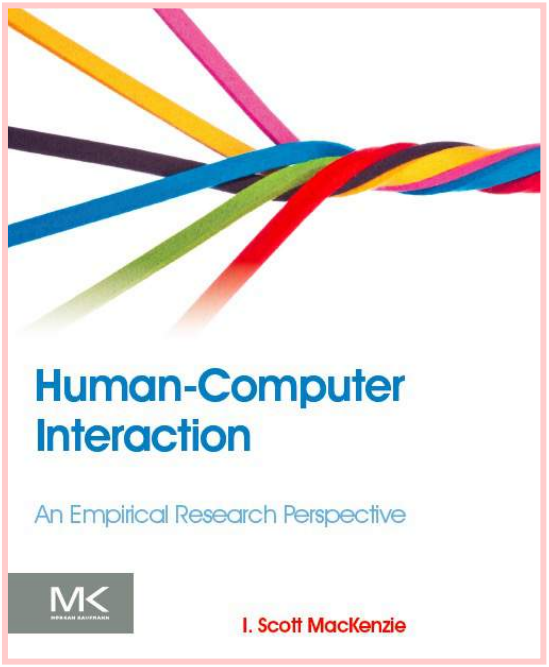
[1] MacKenzie, I. S. (2009). The one-key challenge: Searching for an efficient one-key text entry method. *Proc ASSETS 2009*, 91-98, New York: ACM.

# Tip #4 – Think Inside The Box

- Think *outside* the box → dispense with accepted beliefs and assumptions (in the box), and think in a way that assumes nothing and challenges everything
- No need
- Think *inside* the box: just get on with your day; but at every juncture, every interaction, think and question
- Our everyday foibles are fertile ground for research topics



# Thank You


Human-Computer Interaction
An Empirical Research Perspective
I. Scott MacKenzie